TSP
Entities
Nodes
I = {1,2,..., i, ...} (By default, I*= I, 1 is the start node)
Cities to visit
Customers to visit
Patients to visit
Edges
E = {ij for all i, j in I and i not j}
Flights
Roads
Rails
Nodes must be visited
A set of nodes must be visited
I^K subset of I
Tag color coding:
yellow for family-holiday instance
red for last-mile delivery instance
green for home-care instance
Default values:
Set of customers must be visited
Set of patients must be visited
Start node fixed
Regular start node
i = 1 for the start node, I* = I
Special start node
i = 1 for the start node, I* = I \backslash {1}
Home is the start node
Pharmarcy is the start node
Depot is the start node
Parameters
Travel costs
c_ij, for ij in E
Airfare
Vehicle cost
Travel times
t_ij for ij in E
Travel time
Service time
s_i for i in I^*
Servicing customers
Examining patients
Penalties
Punctuality
Earliness penalty
p_i^E for i in I*
Arriving too early
Lateness penalty
p_i^L for i in I*
Arriving too late
Missing nodes
p_i^M for i in I*
Not visiting cities
Not servicing customers
Not visiting patients
Values
w_i for i in I*
Attractions
Revenue
Urgency
Budget
Financial limit
b_max
Budget for holiday
Budget for travel cost
Time limit
t_max, t_start (default t_start = 0)
Maximum travel/working time
Time windows
[e_i, l_i] for i in I*
Customer availability
Patient availability/schedule
Objectives
Minimization
Minimizing total travel cost
sum_{ij in E} c_ij x_ij
Total travel cost
Minimizing total travel time
sum_{ij in E} t_ij x_ij
Total travel time
Minimizing total penalty
Penalizing unpunctuality
sum_{i in I*} (p_i^E z_i^E + p_i^L z_i^L)
Visiting customers early or late
Visiting patients early or late
Penalizing missing nodes
sum_{i in I*} p_i^M (1-y_i)
Missing cities
Missing customers
Missing patients
Minimizing total costs including penalty
sum_{ij in E} c_ij x_ij + sum_{i in I*} (p_i^E z_i^E + p_i^L z_i^L) + sum_{i in I*} p_i^M (1-y_i)
Maximization
Maximizing total value
sum_{i in I*} w_i y_i
Total value of joy
Total revenue
Maximizing total profit
sum_{i in I*} (w_i y_i - p_i^M (1-y_i))
Total profit
Decision variables
Nodes visited
y_i = 0/1 for i in I
Visiting cities
Visiting customers
Visiting patients
Edge traveled
x_ij = 0/1 for ij in E
Flight from city to another city
Travel from customer to another customer
Travel from patient to another patient
Arrival time (if time windows are given)
a_i >= 0 for i in I
Arrival time at city
Arrival time at customer
Arrival time at patient
Early arrival (only if a_i is defined)
z_i^E = 0/1 for i in I*
Arriving at city earlier than scheduled
Arriving at customer earlier than scheduled
Arriving at patient earlier than scheduled
Late arrival (only if a_i is defined)
z_i^L = 0/1 for i in I*
Arriving at city later than scheduled
Arriving at customer later than scheduled
Arriving at patient later than scheduled
Position along route
u_i = integer for the position of node i along the route
Constraints
Nodes must be visited
A single starting node (depot)
y_0 = 1
Home
Depot
Medical center
A set of nodes
y_i = 1 for i in I^K
Cities must be visited
Customers must be visited
Patients must be visited
Time window satisfaction
No violation (hard constraint)
e_i <= a_i <= l_i for i in I*
Must visit customers within their nominated time windows
Must visit patients within their nominated time windows
Penalizing violation (soft constraint)
Earliness
M z_i^E >= e_i - a_i for i in I*
May visit customers earlier than their nominated time windows with penalty
May visit patients earlier than their nominated time windows with penalty
Lateness
M z_i^L >= a_i - l_i for i in I*
May visit customers later than their nominated time windows with penalty
May visit patients later than their nominated time windows with penalty
Budget
Financial limit
sum_{ij in E} c_ij x_ij <= b_max
Total expense no more than the budget
Time limit
sum_{ij in E} t_ij x_ij + sum_{i in I*} s_i y_i <= t_max - t_start
Total working time no more than the maximum
Leaving customers no later than the maximum time
Leaving patients no later than the maximum time
Flow balance
sum_{i in I} x_ij - sum_{i in I} x_ji = 0 for j in I
Returning home
Returning to depot
Returning to hospital
Visited nodes
y_i >= sum_{j in I} x_ij for i in I
Each node is visited at most one time by any vehicle (if y_i not defined)
sum_{j in I} x_ij <= 1 for i in I
Arrival time
Travel bound
a_j >= sum_{i in I} x_ij (a_j + s_j + t_ij) for j in I*
Lower bound
a_i >= t_start for all i in I
Subtour elimination
DFJ
MTZ
u_i+1-u_i <= M(1-x_ij) , for i, j in I (M is a big number)
Loop (default is closed loop)
Open loop (start node becomes a dummy node)
c_i1 = 0, t_i1 = 0